

TESTING NEEDS OF AUTOMOTIVE INDUSTRY

From in-vehicle communication to fuel efficiency and regulatory norms, every automotive innovation is becoming more complex and highly integrated. It implies that test and applications engineers must be innovative and resourceful in approaching every test for better design, shorter time-to-market and lower cost



Automotive testing (Courtesy: National Instruments)

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When you build a system that integrates many small systems, the biggest challenge is effective communication between all the smaller systems to make the big integrated system work as one. Today's automobiles consist of various inter-connect buses to communicate between sub-systems. To test these systems, you need tools and solutions that support serial communication standards.

An evident change which the automotive industry has gone through over the years is the use of electronics

in automobiles. "If you see a new-generation car, the use of electronics is conspicuous right from the security and vehicle access system to the driver interface, powertrain, safety and enticing infotainment and telematics," says Ashish Jain, applications engineer, Agilent Technologies.

Test and measurement tools have been used for many years now in the design and testing of engine control systems, entertainment systems and safety systems. "What is now emerging as a critical area is energy conservation and power management," adds Evan Sun, technical marketing

manager, Tektronix. Agreeing to this Prabhakaran P.V., business unit head-MDTS from TTL Technologies, shares, "With the increasing commercial and regulatory pressure for safer and environment-friendly vehicles, most of the automotive OEMs are looking for more efficient and green designs to position themselves on top."

Rajesh Pathak, advanced specialist-technology, Delphi, explains, "An automotive controller has to undergo many tests in sub-systems as well as integrated environment. Some of the tests include vibration analysis, crash testing, accelerated tests, temperature test, EMI/EMC test, safety standard requirement, emission test and structural tests. Apart from these, there are non-electronics related tests like oil test, salt test, water test, soak test, sand test and drop test."

"The test and measuring solutions have to be versatile, adaptable and exhibit excellent forward/reverse compatibility to cater to an extremely varied nature of testing requirements of the automotive industry without affecting the integrity of the test methods," says Tapesh Gautam, assistant manager, product support & application, Rohde & Schwarz.

Commercial off-the-shelf equipment

Automobiles, though specialised applications, are basically an integration of many sub-systems. To test these systems, there are several commercial